

Dynamic Access Control and Privacy Compliance Strategies in a Zero-Trust Data Security Architecture

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Abstract: Artificial Intelligence (AI) and intelligent algorithms are revolutionizing data security governance by transforming access control from a static authorization process to a dynamic risk assessment approach. This paper proposes dynamic access control and privacy compliance strategies within a zero-trust data security architecture, to overcome the problems of unauthorized access, abnormal calls and lack of privacy compliance of cloud platforms, multi-device access, and cross-domain data flows. It does this by creating an integrated mechanism that includes identity authentication, user behavior modeling, multi-dimensional trust score evaluation, risk-aware decision-making, and compliance auditing, enabling continuous verification of access requests and dynamically adjusting permissions. Experimental results demonstrate that the access decision accuracy rate is 92.8%, the average F1 score of anomalous access is 0.912, the compliance audit coverage has been raised to 96.4% and the average access response time in 1000 concurrent connections is 146.3ms, which shows that this method has good security, compliance, and engineering applicability.

Keywords: Zero-Trust Architecture; Dynamic Access Control; Privacy Compliance; Data Security

I. INTRODUCTION

Data security governance has evolved from a static authorization model to continuous access control, based on identity, behavior, environment, and risk status, with the help of AI and intelligent algorithms. The advent of cloud platforms, multi-device access and cross-domain data flows have put the success of the traditional perimeter defense methods in jeopardy, leaving account abuse, unauthorized access, the exposure of sensitive data and the absence of privacy compliance to be the critical challenges to solve for a Zero Trust Data security architecture.

Wang et al. proposed a zero-trust dynamic access control method tailored to cloud computing scenarios, demonstrating that dynamic authorization can enhance the security of cloud resource access [1]; Chinni evaluated adaptive access policies in modern cybersecurity environments, providing a reference

for the flexible adjustment of permissions [2]; Chaturvedi et al. elucidated the compliance value of zero-trust architectures from the perspective of digital privacy protection in healthcare [3]; Potluri constructed an identity and access management framework for cross-cloud federated networks, reinforcing the importance of trustworthy identity management in multi-cloud environments [4]; Anbazhagan introduced intelligent models into enterprise data protection and infrastructure management, expanding the application boundaries of the zero-trust security model [5]; Ahmadi analyzed the application challenges and development opportunities of zero-trust architectures in cloud networks, pointing out that constraints still exist in policy coordination and engineering implementation [6].

Therefore, mechanisms of identity authentication, trust evaluation, risk-sensitive access decision, and privacy compliance audit can be built on the zero-trust data security architecture to obtain dynamic control, anomaly detection, and compliance audit during the whole data access process, which provides a systematic approach to improve data security protection capability and data privacy governance level.

II. DESIGN OF THE ZERO-TRUST DATA SECURITY ARCHITECTURE

A. Overall System Architecture

Its main components are the data resource layer, identity authentication layer, trust assessment layer, access control layer, privacy compliance audit layer and security management layer. Once an access request is made by either a user or an application, the system first identifies the user and the status of the device calling the access request [7]. It subsequently measures a trust score from the integration of behavioral characteristics, access environment and data sensitivity levels and takes a decision about accessing the data based on the trust score. The audit module records operation logs, permission changes and abnormal behaviors during the access process, and establishes a traceable and feedback security loop. The overall system structure is shown in Figure 1.

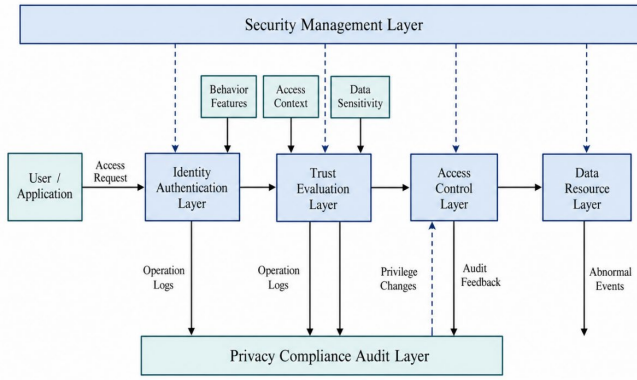


Figure 1. Overall System Architecture

B. Identity Authentication and Trust Assessment Module

The system checks the identity of the subject by performing multi-factor authentication, device fingerprinting, token verification, and credential comparison, when a user or application makes a request. It then determines a detailed trust score, taking into account device status, behavioral anomalies, access environment and history [8]. The multi-dimensional weighted approach is used to represent the trust score:

$$T = \sum_{i=1}^n w_i x_i \quad (1)$$

where T is the comprehensive trust score; x_i is the i -th trust evaluation metric, such as identity reliability, device security, or behavioral stability; w_i is the corresponding weight; and n is the number of metrics [9].

To account for the impact of anomalous behavior, a risk adjustment factor is introduced:

$$T' = T(1 - R) \quad (2)$$

where T' is the adjusted trust value; R is the risk coefficient, with a range of [0,1]. When R increases, the trust value decreases, and the system will trigger secondary authentication, access downgrading, or blocking policies [10].

To enforce the Zero-Trust principle of continuous verification, trust is not permanent. The final trust score incorporates a temporal decay function: $T_{\text{final}} = T_{\text{adjusted}} \cdot e^{-\lambda \Delta t}$, where Δt is the time elapsed since the last trust evaluation, and λ is a configurable decay coefficient. In our implementation, $\lambda = 1/300$. Furthermore, any privilege escalation attempts or anomalous event immediately sets $\Delta t \rightarrow \infty$, forcing a trust collapse and re-authentication.

C. Data Access Control Module

The Data Access Control Module dynamically produces authorization results based on authentication results, trust scores, data sensitivity levels, access operation types. With an access request, the system will first compare the user's role with the attributes of the resource and then, based on the real-time risk status, it will either grant or deny access or initiate

secondary verification. The access decision function can be expressed as:

$$D = f(T', S, O, E) \quad (3)$$

where D is the access decision result, O is the access operation type, such as read, download, or modify; and E is the access environment factor [11]. The permission level is calculated as follows:

$$P = \alpha T' - \beta S - \gamma R_o \quad (4)$$

where P is the permission score; α , β , γ is the weighting coefficient; R_o is the operational risk value [12]. When P exceeds the set threshold, access is granted; when it falls below the threshold, de-identification, read-only, downgrading, or blocking policies are enforced, thereby implementing the principle of least privilege.

D. Privacy Compliance Audit Module

The system compares access behavior against compliance rules to identify issues such as unauthorized calls, out-of-scope usage, or missing audits. The compliance score can be expressed as:

$$C = \frac{N_c}{N_t} \times 100\% \quad (5)$$

where C is the compliance coverage rate; N_c is the number of access events that meet compliance requirements; and N_t is the total number of access events [13]. The compliance risk value can be expressed as:

$$V = \lambda_1 A + \lambda_2 M + \lambda_3 L \quad (6)$$

where V is the violation risk value; A is the degree of unauthorized access; M is the degree of sensitive data exposure; L is the degree of missing logs; and λ_1 , λ_2 , λ_3 is the weighting coefficient. If V exceeds the threshold, the system will trigger alerts, audit trails, and access policy corrections [14].

III. CONSTRUCTION OF THE DYNAMIC ACCESS CONTROL MODEL

A. Modeling User Behavior Characteristics

User behavior feature modeling is the foundation of dynamic access control, with the goal of transforming discrete access logs into computable behavior vectors. The system extracts features from dimensions such as login time, access frequency, resource type, operation method, download volume, device switching, geographic location, and number of failed authentication attempts to form a description of user behavior:

$$X_u = (x_1, x_2, \dots, x_n) \quad (7)$$

where X_u is the behavioral feature vector of user u ; x_i is the behavioral feature of class i ; and n is the number of feature dimensions [15]. To avoid overly permissive or restrictive access during the cold start phase, new users are assigned a provisional trust score of 0.4 and granted only minimal, read-only access to non-sensitive data. A behavioral baseline is

constructed from the first 20 access attempts, after which the full trust scoring and dynamic adjustment mechanisms take effect. To measure the degree of deviation between current behavior and historical baselines, a behavioral deviation value is introduced:

$$B = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}_i| \quad (8)$$

where B is the behavioral deviation value; x_i is the current access feature value; $\bar{x}_i$ is the mean of the user's historical normal behavior [16]. The larger B is, the more abnormal the access behavior is, and the system will elevate the risk level and restrict permissions. The feature modeling process is shown in Figure 2.

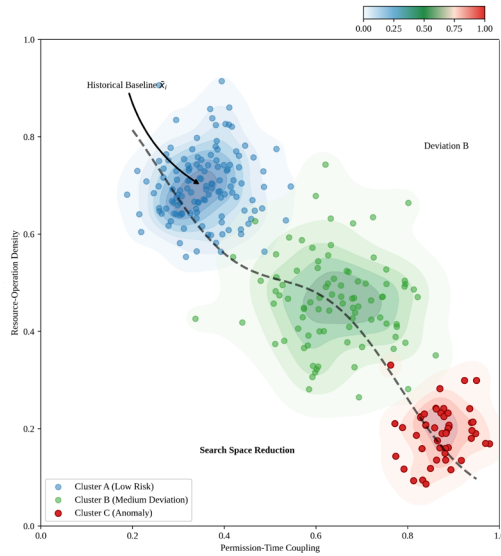


Figure 2. Feature Modeling Mapping Diagram

B. Method for Calculating Multidimensional Trust Scores

The system integrates factors such as identity credibility, device security status, behavioral anomalies, access environment, data sensitivity level, and historical violation records to construct a dynamic scoring model:

$$T_s = \sum_{i=1}^m w_i q_i - \delta B - \mu H \quad (9)$$

where T_s is the comprehensive trust score; q_i is the i th trust metric; w_i is the metric weight; B is the behavioral deviation value; H is the historical risk record; and δ , μ is the penalty coefficient. A higher score indicates lower access risk and grants higher privileges; when the score decreases, the system automatically triggers privilege downgrading, secondary authentication, or access blocking.

C. Risk-Aware Access Decision Mechanism

When asked, the system does not just look at what kind of user it is, but at a baseline of the user's past behavior, current device status, access time, network location, operation type, etc. If access behavior matches historical access patterns, data

sensitivity is low, the system allows normal access; if there are any indicators of access risk, such as accessing from a remote location, frequent downloads, access from an unauthorized device, or access from a suspect device, the system raises the risk level and uses features like two-factor authentication, read-only access, field redaction, access downgrade, or access blocking.

D. Design of the Dynamic Permission Adjustment Algorithm

The dynamic permission adjustment algorithm automatically updates permission status based on the access subject's real-time trust score, risk level, and data sensitivity. Upon receiving each access request, the system first reads the current trust value and risk factors, then determines whether to maintain, lower, or revoke permissions. The permission adjustment function can be expressed as:

$$P_t = P_{t-1} + \eta(T_s - \theta) - \rho R \quad (10)$$

where P_t is the current permission level; P_{t-1} is the permission level at the previous time step; η is the trust adjustment coefficient; T_s is the comprehensive trust score; θ is the trust threshold; ρ is the risk penalty coefficient; and R is the current risk level. If P_t exceeds the authorization threshold, the system grants access; if it falls below the threshold, it triggers a downgrade, read-only access, data masking, or access blocking strategy, thereby achieving dynamic convergence of permissions in response to risk changes.

IV. METHODS FOR OPTIMIZING PRIVACY COMPLIANCE POLICIES

These methods span the entire lifecycle of data collection, storage, access, transmission, sharing and destruction, and are designed to help to conquer problems like a lack of identification of sensitive data, unclear demarcations of who has permission to use the data or what they do with it and a delay in feedback from audit. Then, it defines protection levels for each type of data—identifying data, core business data, and operational data—according to data contents, business purpose, and data sensitivity, and it then configures different protection control strategies based on the combination of the identity of the entity being accessed, business purpose, and risk score. The system uses encrypted storage, field redaction, least-privilege authorization and access approval mechanisms for highly-sensitive data, while dynamic authorization and log tracking are used for routine management of general data. The audit results can be used for a later refinement of the compliance rules: If it is detected that unusual calls, accesses outside the scope or missing logs were made, the intensity of the audits is increased and access policies are optimized, creating a constantly improving privacy protection mechanism.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Environment Configuration

The hardware specification is an Intel Core i9 processor, 32 GB of RAM, and NVIDIA RTX 3080 GPU. This system is running on Ubuntu 20.04, programming language of development is Python 3.9. The user information, access logs and audit information are stored in MySQL and MongoDB. Experimental data was developed by simulating access

scenarios on enterprise data, using representative behavior like regular logins, remote access, frequent downloads, unauthorized logins, queries for sensitive data and unusual device connections. The data is split into a training, validation and test set in a 7:2:1 ratio. Finally, evaluation metrics, such as access decision accuracy, abnormal access detection rate, false positive rate, F1-score, average response time, compliance coverage, and audit log completeness, are employed to thoroughly assess a system’s security, compliance, and operational performance.

B. Performance Analysis of Dynamic Access Control

To evaluate the actual effectiveness of the dynamic access control mechanism, we compare RBAC, ABAC, rule-based risk control, and the method proposed in this paper, focusing on the acceptance of legitimate access, blocking of unauthorized access, dynamic de-privileging, and access decision accuracy.

TABLE I. COMPARISON OF DYNAMIC ACCESS CONTROL PERFORMANCE

Method	Normal Access Pass Rate / %	Unauthorized Access Blocking Rate / %	Dynamic Privilege Downgrade Success Rate / %	Access Decision Accuracy / %
RBAC Static Role-Based Control	91.6	82.4	68.7	84.9
ABAC Attribute-Based Control	92.8	86.1	74.5	87.6
Rule-Based Risk Control	90.9	89.3	81.2	89.1
Proposed Method	94.2	93.7	88.6	92.8

From the results in Table I, it can be seen that the accuracy of access decision of the access control method proposed in this paper reaches 92.8%, which is 7.9% higher than static RBAC role-based access control; the blocking success rate of unauthorized access reaches 93.7%, and the success rate of dynamic de-privileging reaches 88.6%. This means that the method can update permissions in real-time for changes in risk, and lessen the risk of unauthorized access with static authorizations.

C. Evaluation of Anomalous Access Detection Performance

To evaluate the system’s ability to identify anomalous access behavior, we conducted tests using typical risk scenarios such as remote logins, high-frequency downloads, access outside working hours, connections from unusual devices, and unauthorized access attempts. The results are shown in Table II.

TABLE II. DETECTION PERFORMANCE IN DIFFERENT ABNORMAL ACCESS SCENARIOS

Abnormal Access Type	Precision / %	Recall / %	F1-score	False Alarm Rate / %
Remote Login from Unusual Location	92.5	90.8	0.916	5.9
High-Frequency Download	91.7	89.6	0.906	6.4
Non-Working-Hour Access	89.8	87.3	0.885	7.1
Abnormal Device Access	93.1	91.4	0.922	5.6

Unauthorized Access Attempt	94.0	92.7	0.933	4.8
Average	92.2	90.4	0.912	6.0

The average Precision and Recall are 92.2 % and 90.4 % respectively while the average F1-score is 0.912, as shown in Table II. Of these, the identification performance of handwriting for unauthorized access attempts is the highest, F1 score of 0.933, and false positive rate as low as 4.8%, while the F1 score of access outside the working hours is 0.885, which shows that handwriting in complex business scenes still has an influence on judgment threshold (See Figure 3).

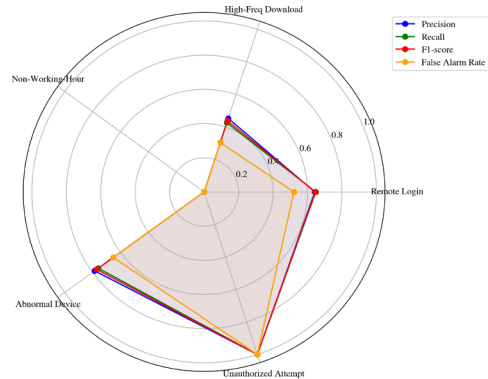


Figure 3. Comparison of abnormal access scenario detection

It should be noted that the average false positive rate of 6.0% could, in a production environment, trigger unnecessary step-up verifications or temporary restrictions on legitimate users, potentially leading to access fatigue. Our framework mitigates this operational impact through a graded response strategy: ambiguous risk events do not lead to outright blocking, but rather invoke secondary authentication, read-only mode, or field-level redaction. This ensures that even when a false positive occurs, the user retains partial access and can resume full privileges upon re-verification, thus balancing stringent security with operational continuity. Furthermore, the risk sensitivity threshold can be tuned per deployment scenario to shift the balance as required.

D. Analysis of Privacy Compliance Strategy Effectiveness

The effectiveness of privacy compliance strategy optimization is analyzed through a comparative assessment of sensitive data identification, data masking execution, audit coverage, violation interception, log integrity, and compliance report generation, as shown in Table III.

TABLE III. COMPARISON OF RESULTS BEFORE AND AFTER PRIVACY COMPLIANCE POLICY OPTIMIZATION

Indicator	Before Optimization / %	After Optimization / %	Improvement / Percentage Points
Sensitive Data Identification Rate	88.4	95.1	6.7
Desensitization Strategy Execution Rate	84.7	93.6	8.9
Compliance Audit Coverage Rate	86.2	96.4	10.2
Violation Access Blocking Rate	83.9	92.8	8.9

Audit Log Integrity Rate	90.5	97.2	6.7
Compliance Report Generation Accuracy	87.6	94.5	6.9

As shown in Table III, after strategy optimization, compliance audit coverage increased from 86.2% to 96.4%, a rise of 10.2 percentage points; the data masking strategy execution rate increased from 84.7% to 93.6%, and the rate of blocked unauthorized access reached 92.8%. The data indicates that the optimized compliance strategy can more effectively regulate the process of accessing sensitive data.

E. System Performance Analysis

The operational overhead of the Zero Trust architecture during continuous verification, dynamic scoring, and audit logging processes was evaluated by setting different scales of concurrent requests to test response time, policy matching efficiency, and system throughput. The results are shown in Table IV.

TABLE IV. SYSTEM PERFORMANCE COMPARISON

Number of Concurrent Requests	Average Response Time / ms	Policy Matching Time / ms	Audit Write Latency / ms	CPU Utilization / %	Throughput / req·s ⁻¹
100	46.8	11.2	8.5	28.6	94.7
300	63.5	15.9	10.7	41.3	274.2
500	82.4	21.6	13.9	56.8	438.5
800	118.7	31.4	19.5	71.2	651.6
1000	146.3	39.8	25.1	79.5	768.4

As shown in Table IV, when the number of concurrent requests increased from 100 to 1,000, the average response time rose from 46.8 ms to 146.3 ms, CPU utilization increased to 79.5%, and throughput reached 768.4 req·s⁻¹. Overall, the system maintained relatively stable access control efficiency under high-concurrency conditions (See Figure 4).

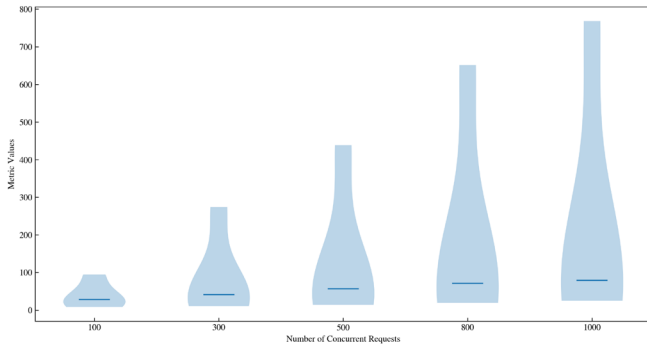


Figure 4. System Performance Comparison

Fig. 4 shows near-linear growth in response time and CPU usage up to 1,000 concurrent requests, with throughput reaching 768.4 req/s. The risk-aware engine evaluates trust in O(d) time per request and uses in-memory hash-based policy matching, keeping latency stable as attributes grow. Each session consumes 1.8 MB memory, allowing a single 32 GB node to support over 15,000 concurrent sessions. For higher scale, the stateless Policy Decision Point can be horizontally scaled behind a load balancer.

VI. CONCLUSION

The Zero Trust data security architecture brings into a single security loop a continuous verification of data access, identity authentication, trust assessment, and dynamic authorization, while ensuring compliance with privacy regulations. The combination of multi-dimensional trust scoring, behavioral anomaly modeling and dynamic permission adjustment mechanisms will further help to detect and prevent unauthorized access, abnormal device connections, and sensitive data access; while integration of privacy compliance policies increases the traceability, standardization, and clarity of accountability boundaries in data processing. Experimental evaluation indicates this architecture obtains good performance on access decision correctness, the effectiveness of anomaly detection, the coverage of compliance and the efficiency of system response. These limitations include the model still requiring high quality logs, stable baseline behaviors and rule weights, while policy synchronization and performance overhead in large-scale heterogeneous environments still require further study. Going forward, continued research efforts may involve cross-domain data flow, automated policy inference, and collaboration for privacy-aware computing to further improve the adaptability and engineering reliability of the Zero Trust data security framework.

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